

Institute of Actuaries of India

Subject ST4 – Pension & Other Employee Benefits

April 2016 Examination

INDICATIVE SOLUTION

Introduction:-

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) The impact of disclosure on security**

- Will not directly improve the security of members' benefits
- Could indirectly increase security as follows:
- Issuing sufficient and relevant information would allow members to judge whether the Scheme was being run on a sound financial basis
- and in accordance with the trust deed and scheme rules and that robust administrative procedures were in place
- It may provide an early warning system to alert members to potential problems
- Members may then put pressure on the scheme sponsors, trustees or regulators to ensure that any problems are quickly rectified
- Alternatively members may choose to take a transfer value of accrued benefits if they feel that the security is inadequate

[3]

ii) Six pieces of key information to improve security

- Details of benefits to be provided, eligibility to join the scheme, any options available and contribution requirements of membership.
- Details of the scheme constitution, legal and tax position.
- Details of the financing of the scheme on various bases, the contribution obligations, funding objectives
- Investment policy, current assets held and investment performance.
- Details of who is involved in running the scheme and what their role is
- Details of their actions and how their performance is measured.
- Details of the scheme complaints procedure.

[3]

iii)**Advantages**

- To help members to understand the financial aspects of the scheme and their benefit entitlements and options available.
- To alert members to potential problems with the scheme and they can therefore take the actions outlined in part (i) above.
- To make the running of the scheme more transparent.
- This may improve the confidence of members in the running of the scheme.

Disadvantages

- Members may have a lack of understanding of, or interest in, the material provided.

Error! Unknown document property name. 2 Error! Unknown document property name. 14

- There may be too much information available so members do not pick up the important information.
- May create expectations which will not be borne out in practice.
- There will be extra administration and communication costs.

[3]

iv) Other methods of regulation

- Appropriate funding with regular funding checks or a requirement for a minimum level of funding
- Sensible investment with restrictions on assets held and/or funds held separately from the employer's assets.
- Some/all the individuals running the scheme to be elected and/or independent of the employer.
- Use of insurance to provide benefits members if the scheme defaults
- A central fund to meet benefits should the scheme fail, eg the Central Discontinuance Fund.
- A requirement that all parties involved in running the scheme are qualified and trained.
- An independent body could be set up to oversee the running schemes of pension schemes

[3]

[12 Marks]**Solution 2:****i) Reasons why costs may be rising to State**

The demographics in respect of the costs of state provisions are:

- increasing longevity ...
- ... worsened if combined with a falling birth rate, and
- birth booms in the past.

This is exacerbated if individuals have a shorter working lifetime due to:

- a lower proportion of the working population to the retired population
- more time spent in higher education
- career breaks and
- earlier retirement

The extent of the impact on costs will depend upon whether the scheme is financed by PAYG or a funded approach.

The impact on costs will be more noticeable if the scheme is financed by a PAYG method.

Error! Unknown document property name. 3 Error! Unknown document property name. 14

Also need to consider whether contributions are earnings-linked or not.

If the scheme is funded then the rise in costs may reflect poor investment returns earned on the fund.

Costs may increase if inflation-linked benefits are offered and inflation has been higher than expected.

[5]

ii) Reasons why Lack of private provision

The lack of private provision may be due to:

- concerns about the security and risks of such provision
- a lack of knowledge about the importance of making provision
- a lack of provision leading to a lack of competition in the market and high costs, further reducing private provision
- apathy and the belief that the State will provide sufficient benefits
- low wages, so no spare money to make savings
- a lack of tax and other incentives (relative to other savings vehicles)
- a lack of incentives for employers or other providers to offer schemes
- lack of paternalism by employers, eg if the workforce is highly mobile
- over-regulation of occupational schemes, making employers less inclined to offer them, due to the time and costs associated with compliance
- over-regulation of private providers, eg on selling, solvency margins etc
- high charges (for personal pensions or occupational schemes).

[4]

iii) Advantages of the proposal of National Pension Savings Scheme

- if combined with reducing State benefits it can control State costs
- general inertia means many people may remain in the scheme even though an opt-out option is offered, thus increasing provision and overall savings
- the opt-out option is more politically acceptable than if the arrangement is compulsory
- easily copes with change of employment
- requiring employer contributions should help ensure a non-trivial amount of money is saved, leading to meaningful benefits
- requiring only modest employer contributions should not be too onerous a burden on employers
- State operated and therefore in a developed country expected to be secure
- a funded arrangement so may have positive impact on investment markets, creates demand for investments

- offering investment choice enables individuals to tailor investments to their risk appetite and circumstances
- money purchase in nature and therefore transparent and easy to understand
- money purchase, this may allow flexibility over the form and timing of the benefit
- funded and therefore likely to be more secure
- there should be benefits from economies of scale, *ie* lower administration and commission costs

Disadvantages of the proposal of National Pension Savings Scheme

- does not cover long-term unemployed, long-term carers, sickness
- not compulsory, and so doesn't ensure employed individuals have private provision
- unpopular as it may "appear" to be compulsory
- may have a negative impact on existing private provision, eg will existing final salary schemes be more likely to close?
- does not guarantee the pension since money purchase in nature and the investment risk and longevity risk falls on the individual ...
- ... eg there is a risk of poor investment returns over the period to retirement or a fall in value at retirement
- ... eg there is a risk of the conversion terms at retirement being worse than expected due to longevity improvements
- individuals may feel they do not have sufficient knowledge to make investment choice, or may just make inappropriate choices ...
- ... leading to the need for a default option or individuals paying for investment advice
- contributions made may not be sufficient to provide an adequate pension
- may be unpopular with employers as required to contribute
- there may be a need to offer some guarantee or underpin to the pension, with associated costs
- if the State offers tax incentives as encouragement then there is an associated cost
- a significant change to the system with associated costs, eg education will be required and a new administration system
- if a PAYG approach to financing has been adopted in the existing State system then the current and future generation must also contribute to benefit costs for existing pensioners
- additional regulation will be needed
- there may not be a large enough market to support the buyout of benefits at retirement
- may not lead to a higher level of savings, just a change of savings vehicle.

[11]

[20 Marks]

Solution 3:**i) Direct Property Investment****Advantages**

- Direct property investment has in the past achieved higher real return in the long term than government or corporate bonds and cash.
- Direct property investment provides diversification from equities when real returns are required.
- Direct property investment generally produces less volatile returns than equities
- Direct property investment can produce a high running yield with (generally) rents only increasing
- Direct property is expected to achieve real returns in the long term and can be held to back real liabilities.
- The relative advantage of this depends on the type of benefits provided.

Disadvantages

- Direct property has lower expected returns than equities.
- Direct property investment can have poor liquidity and poor marketability.
- Property investments can have a large unit size so if the scheme has little surplus or is small then direct property investment may be difficult.
- Over time a closed scheme will reduce in size and therefore insufficient funds may be available to purchase properties.
- There can be high costs, eg dealing costs, management fees, valuation fees
- Income is potentially volatile, in particular if a property is not occupied

[5]

ii) Characteristics of Projected Unit Method (PUM) and Attained Age Method (AAM)**Projected Unit Method (PUM)**

- actuarial liability (AL) is present value of accrued benefits based on projected salary
- assets = AL if standard contribution rate (SCR) paid throughout and assumptions borne out in practice
- SCR is present value of benefits accruing in control period based on projected salary divided by present value of salary in control period
- SCR stable if stable age/ sex salary profile, which is unlikely here as closed scheme
- control period usually taken as a year

Attained Age Method (AAM)

- AL same as under PUM
- AL is not maintained by payment of SCR
- SCR is present value of all future benefits based on projected salary divided by present value of projected future salary
- SCR under AAM is higher than under PUM if term to retirement greater than one year
- overstates contribution requirements for open scheme, as no allowance for new entrants who are usually younger than average which will lead to a surplus
- appropriate for closed scheme if assumptions borne out

[5]

iii) Formulae for PUM and AAM SCR

The PUM SCR for the scheme, assuming 1-year control period can be expressed as:

$$\text{PUM SCR} = \frac{\frac{1}{A} \times \sum_{j=1}^n S_j \times \left(\frac{1+e}{1+i} \right)^{R-x_j} \times a'_R}{\sum_{j=1}^n S_j \times a_1}$$

The AAM SCR for the scheme can be expressed as :

$$\text{AAM SCR} = \frac{\frac{1}{A} \times \sum_{j=1}^n S_j \times (R-x_j) \times \left(\frac{1+e}{1+i} \right)^{R-x_j} \times a'_R}{\sum_{j=1}^n S_j \times a_{R-x_j}}$$

where:

- R = assumed retirement age
 n = number of active members at the valuation date
 x_j = age of the j th member at the date of the valuation
 S_j = salary (or earnings) of the j th member at the date of the valuation
 A = rate of pension accrual
 e = assumed annual earnings (or salary) growth
 i = pre-retirement discount rate of interest
 a_1 = an annuity to determine the present value of earnings over the next year (allowing for the timing of contributions and salary increases)
 a'_R = value of an annuity payable from age R

(allowing for any contingent spouse's pension and pension increases)
 a'_{R-x} = an annuity to determine the present value of all future earnings

[4]

iv) Relationship between AAM SCR and PUM SCR

The AAM SCR is broadly equal to the PUM SCR multiplied by the expected future working lifetime of the active members and divided by the annuity to value the present value of all future earnings.

This ignores ail as this is an approximation only Thus if F is the average future working lifetime then we can estimate the AAM SCR as:

$$\text{AAM SCR} = \text{PUM SCR} \times F \times \frac{a_{\overline{1}}^{i-e}}{a_{\overline{F}}^{i-e}}$$

It may be necessary to also estimate F using the weighted average age and expected retirement age of the active membership.

[3]

[17 Marks]

Solution 4:

i) Trustees to consider

- Assessment of employer covenant i.e. the ability and willingness of sponsor to pay sufficient contributions
- Current and future investment strategy
- The current funding objective and last times' valuation assumptions and results
- Funding method
- Profile of liabilities and extent of asset matching
- Historical salary data in the country, industry or company
- Historical mortality data split by category e.g. factory, administration and management
- Historical scheme specific experience e.g. leavers, ill health retirements, proportion married etc split by category
- Information from the sponsor regarding any planned activity e.g. redundancy, withdrawal and early retirement policy
- Current market data and government policy statements / intentions
- Any Legislative requirements e.g. use of standard / fixed assumptions
- Assessment of whether past data / information is relevant for the future

Error! Unknown document property name. 8 Error! Unknown document property name. 14

- And is it statistically credible?
- View of future economy
- Allowance for discretionary benefits

[5]

ii) Key Financial and Demographic Assumptions

I. Financial assumptions

Prudence

- Degree of prudence should depend on employer's covenant
- Not necessarily in every assumption
- Overall level should be considered

Investment Return

- Based on implied market discount rate on assets matching the liabilities.
- Allowance for future asset outperformance in excess of gilt/ bond yield (risk premium)
- Separate assumptions for pre and post retirement may be used
- Or a combined pre & post assumptions reflecting the weighted average of the individual discount rates based on the proportions invested in each asset class
- Could just look at bond yields

Price inflation

- Use long term view of expected price inflation
- Could compare the yield on fixed interest and index linked Gilts / Bonds
- Or Government projections of price inflation

Pension Increases

- Defined in scheme rules or established practice
- If related to price inflation
- Hence base on price inflation assumption derived above
- Allowing for volatility

Salary growth

- Consider promotional increases
- split by works / staff
- And general inflationary increases in salaries (linked to price inflation)
- Discussions with sponsoring employer

II. Demographic assumptions

Mortality in retirement

Error! Unknown document property name. 9 Error! Unknown document property name. 14

- Look at scheme experience of separate homogenous groups eg manual / office based workers
- Determine a suitable base mortality table
- Probably based on standard tables with an appropriate adjustment
- Most important demographic assumptions
- Allow for future improvements in mortality

Other key assumptions

- Allowance for expenses

[8]

iii) Valuation Process

- Valuation is a snap shot at a point in time
- But a long term view is needed
- The sponsor covenant is important for long term security of members' benefits (i.e. strong covenant allows a greater degree of flexibility)
- In volatile market conditions need to show different possible funding plans
- Investigate the sensitivity of results to changes in key assumptions
- And differing market conditions
- Monitoring development becomes more important
- Consider developing triggers for future actions e.g. an updated valuation / extra cash.
- Consider increased explicit margins / prudence
- Discussions between sponsor / trustees are more important to ensure understanding of the impact and sensitivities of the assumptions.
- Consider smoothing the key assumptions (averaging over a period) if market related assumptions are to be used
- Consider if any regulator guidance

[4]

iv) Reasons for reviewing Investment Policy

- The liability structure may have changed significantly. For example:
 - following a takeover or sale where the membership profile has changed significantly
 - or recent benefit changes such as removing the final salary link and giving all active members leaving service benefits
 - or benefit improvements
 - or legislative changes such as providing guaranteed pension increases for all service, equalisation etc.
- The funding position may have changed significantly.

- For example, surplus may have disappeared or deficit increased following a fall in the stock market, higher than expected salary increases etc.
- The investment manager may have significantly underperformed relative to its performance objective.
- For example, the manager has changed its style, staff movements, adopting too much risk.
- Also need to consider whether any Scheme events have impacted on the manager's performance e.g. a large disinvestment at short notice which adversely impacted the manager's performance relative to its performance target.
- Employer covenant may have strengthened (allowing more risk to be taken) or weakened (allowing less risk to be taken).

[3]

v) Investment Strategy Review – General Issues

- The liability profile:
 - nature (proportion of fixed or real liabilities) and term (short/long related to maturity of the scheme)
- The funding position:
 - is the scheme in surplus allowing greater investment freedom or is the scheme in deficit.
- The size of the fund:
 - whether it is increasing, static or decreasing related to maturity of the scheme, whether it is open/closed to new entrants and future accrual.
- The expected cash flow & liquidity requirements.
- Current strategy and expected costs of revising.
- Trustee and sponsors attitude to risk
- The strength of the employer covenant and its long term commitment to funding the pension scheme.
- Requirements of any trust deed & rules.
- Any legislative or taxation constraints.

[3]

vi) Objective of ALM Exercise

- The objective of the ALM is to project future asset and liability cash flows using stochastic and deterministic methods to obtain a range of likely outcomes in order.
- to help in assessing the risk (i.e. probability of shortfalls) and rewards (i.e. probability of surpluses) of holding different investment strategies.

[1]

vii) Four different measures of risk are

- The probability that the Employer contribution rate rises above x% over the next 10 years is less than 5%.
- The probability that the ongoing funding level falls below 75% over the next 10 years is less than 5%.
- The investment strategy required to achieve an ongoing funding level of 100% over the next 10 years with 90% probability.
- The probability of exceeding 120% funding level on an accounting basis over the next 10 years is less than 5%

[2]

viii) Information Required for ALM Exercise

- Trustees' objectives.
- Time period for the projections.
- Funding method, funding assumptions and results (funding level and future contributions) from the actuarial valuation
- Full membership data, asset information required for the actuarial valuation
- Details of terms of any options (e.g. cash commutation, early retirement, transfer values) and guarantees as the timing of the cash flows is important in an ALM.
- Number of simulations to be run – in order to obtain reliable statistical estimates 10,000 simulations will usually be necessary.
- Economic model needed to project cash flows e.g. random walk, wilkie model
- Parameters for the model – expected returns and standard deviations of
- return on each asset class, the degree of correlation between different
- parameters e.g. equity returns and price inflation,
- Initial best estimate assumption for the model – both financial and
- demographic e.g. withdrawal, mortality, early retirements and proportions
- assumed to exercise options.
- These may differ from the funding assumptions which contain margins for prudence.

[3]

ix) Results of ALM Exercise

- Statistics on the distribution (typically expressed as mean, standard deviation, lower and upper quartiles) of possible future valuation results at the end of the projection period for different investment strategies.
- The valuation results can be expressed in different ways e.g. funding level or contribution rate on different bases – ongoing, solvency, accounting.
- A range of sensible investment strategies for a particular set of investment return and risk assumptions often called an “efficient frontier”.

- Statistics on the distribution of the future net (benefit outgo less contributions and investment income) cash flows for each year over the projection period – to assess the likelihood of realisation of assets on unfavourable terms.

[5]

[34 Marks]**Solution 5:****i) How may affect negotiations**

- Need to consider position of schemes
- The funding deficits (or surpluses) may be large relative to the purchase price
- There may be disagreements on the actual size of any pension scheme deficits
- Assessment and understanding of the underlying risks in the pension schemes is needed
- e.g. investment strategy, future contribution requirements, legislative changes
- Trustees acting on behalf of members' security
- Which may conflict with the employer's objectives
- Will need to be aware of Trustee powers in Trust Deed and Rules
- e.g. power to wind up scheme, power to set contributions
- Covenant strength of new combined employer
- There may be legislative and /or pension regulator involvement to consider
- Impact on future Company share price
- Conditions imposed as part of sale e.g. scheme to be de-risked crystallising a larger deficit
- or extra contribution required to remove deficits in the schemes
- Different benefit structures may complicate matters

[5]

ii) Key Issues that Company Y should consider

- Company's primary duty is to its shareholders
- Strategic and business implications are the key drivers
- The pension scheme may act as a deterrent to the purchaser
- Future share prices will factor in the underlying pension position and risk
- May be requests from the Trustees for the schemes to be de-risked
- Or contributions to return to full funding
- Any pension scheme deficits will need to be factored into the price negotiations
- But will be priced differently by different parties perhaps anticipating Trustee reaction / requests
- The size of the deficit may be large in relation to sale price
- Full disclosure of all pension scheme details will be needed

- Including up to date funding positions
- Possible legislative / regulator involvement
- Company should take its own legal and actuarial advice
- Need to understand the future impact on balance sheet and other accounting disclosures
- Company Y may explore the possibility of excluding the Company A pension scheme from the transaction
- Need to consider the practical issues regarding any participation periods
- Whether to harmonise benefits (if different)
- Consider employee relations
- Problems exacerbated by scheme a having lots of former employees as members – little interest to Company Y?

[6]

iii) Key Principles that the Trustees should consider

- Primary consideration will be the impact on the covenant
- How future funding shortfalls will be addressed
- Engage with the Company to perhaps request increased contributions and or any additional security offered
- Or seek to reduce investment risk
- Need Information on any possible restructuring resulting in early retirement / redundancies if they affect the finances of the Schemes
- Preserving or enhancing the security of accrued benefits is a key consideration
- Updated funding positions will be needed
- Understanding the details & powers in Trust Deed & Rules e.g. contributions, wind up etc
- Regulator / Legislative considerations
- Managing potential conflict of interest for some of the Trustees
- Seek legal & actuarial advice at an early stage
- Decisions to be made on combining the schemes or keeping them separate
- Consider admin, trustees, and investments
- Communication with members etc
- Investment issues, liquidity requirements

[6]

[17 Marks]
